Predicting Food-Security Crises with GDELT News Features

Implementation and Results Summary

1. Overview

Pipeline objective: test whether GDELT-derived news features improve district-level food-security crisis prediction (IPC Phase >= 3) beyond autoregressive (AR) baselines alone, across 18 African countries using 353 livelihood zones.

2. Data

IPC assessments: 3,245 district-period observations, 353 livelihood zones, Oct 2020 to Feb 2024. Crisis prevalence: 28.0% (909 of 3,245 at IPC >= 3).

GDELT news features (18 per fold): 9 topic relative-coverage proportions (conflict, displacement, economic, food security, governance, health, humanitarian, weather, other) and 9 corresponding district-normalised z-scores.

3. Model Architecture

Two CatBoost classifiers (5-feature AR-Only vs 23-feature AR+News) are compared per fold.
AR features (5): ipc_lag_1, ipc_persistence_2yr, spatial_lag, ipc_period (cat.), ipc_country (cat.).
AR+News adds 18 GDELT news features to the identical AR feature set.

Both models share identical CatBoost params: depth=6, lr=0.03, 1000 iterations, loss=Logloss, eval=AUC.
No early stopping, no class weighting — identical training protocol for fair null comparison.

4. Evaluation Design

Rolling temporal cross-validation: 7 folds, each stepping forward one IPC period.
~2-year rolling training window, 8-month-ahead test horizon (L=2 IPC periods). Test periods span Feb 2022 to Feb 2024.
Total test-set: 2,148 observations (636 crisis, 1,512 non-crisis).

Primary metric: PR-AUC — chosen because it reflects true-positive detection in the minority class and is not inflated by the large non-crisis majority. ROC-AUC is reported as secondary.

5. Null Shuffle Test

To verify that news features carry genuine signal, all 18 news columns are randomly scrambled across both districts and time periods, breaking any alignment with outcomes while preserving global feature quantities. Both models are fully retrained across 1000 such permutations under identical training protocols, producing a null distribution of PR-AUC values. The real model's score is then ranked against this distribution.
p-values are one-sided empirical (fraction of null permutations >= real statistic).

6. Results

Mean cross-validated performance across 7 folds:

Metric	AR-Only	AR+News	Delta	Null mean ± std (n=1000)	p-value
PR-AUC	0.8346	0.8711	+0.0365	0.8452 ± 0.0120	**0.008 ***
ROC-AUC	0.9069	0.9337	+0.0268	0.9244 ± 0.0061	0.028

* $p < 0.05$ (one-sided empirical, 1000-permutation row+column shuffle test)

7. Fold-Level Breakdown

PR-AUC by fold (AR-Only / AR+News / delta):
- Fold 1 (Feb 2022, n=263): 0.889 / 0.899 / +0.009
- Fold 2 (Jun 2022, n=297): 0.907 / 0.904 / -0.003
- Fold 3 (Oct 2022, n=263): 0.909 / 0.886 / -0.024
- Fold 4 (Feb 2023, n=341): 0.862 / 0.885 / +0.023
- Fold 5 (Jun 2023, n=341): 0.714 / 0.887 / +0.174 <- AR collapses; news rescues
- Fold 6 (Oct 2023, n=343): 0.851 / 0.849 / -0.003
- Fold 7 (Feb 2024, n=300): 0.710 / 0.788 / +0.079

News improves PR-AUC in 4 of 7 folds, degrades it slightly in 3. Fold-to-fold std of delta = 0.069.

8. Interpretation

News features significantly improve crisis prediction over the AR baseline (PR-AUC 0.8346 → 0.8711; $\Delta = +0.0365$, $p = 0.008$). The gain is consistent with a genuine news signal: the null test confirms that randomly scrambled news features cannot replicate it.

The improvement is most pronounced in periods where past crisis history alone is a poor predictor — precisely the situations where early warning systems matter most.
ROC-AUC also improves (0.9069 → 0.9337; $p = 0.028$), though PR-AUC is the more operationally relevant metric given the rarity of crisis observations.